

# Thong (Tom) Huynh

 ThongHHuynh |  Thong Huynh |  Portfolio |  thong.huynh@ontariotechu.net

## EDUCATION & AWARDS

|                         |                                                                                                   |           |
|-------------------------|---------------------------------------------------------------------------------------------------|-----------|
| Ontario Tech University | B.Eng., Mechatronics Engineering (GPA: 4.25/4.3)                                                  | 2024–2028 |
| Competitions            | International Youth Robot Competition 2019 (1st place), MakeX (Innovative Design), FIRST Robotics |           |
| Awards                  | Undergraduate Research Award, STAR Award, Merit Award, President’s List                           |           |

## SKILLS

|                          |                                                                             |
|--------------------------|-----------------------------------------------------------------------------|
| CAD and Design:          | SolidWorks Certified Professional, Siemens NX, 3D-Printing, GD&T, DFM/DFA.  |
| Robotics Hardware:       | NVIDIA Jetson Orin Nano/AGX, Raspberry Pi, LiDAR, IMUs, sensors, actuators. |
| Robotics and Simulation: | ROS 2, Gazebo, NVIDIA Isaac Sim/Lab, Nav2.                                  |
| Programming and Vision:  | Linux, Python, C++, Git, OpenCV, PyTorch, YOLO, RF-DETR, DINO/DINOv2.       |

## EXPERIENCE

Robotics Software Engineer | ABI Ltd.

May 2026 – Present

- Developed high-fidelity digital twins of industrial robotic workcells in **NVIDIA Isaac Sim** to accelerate system validation, reducing on-site commissioning time by 30%.
- Architected an in-house, two-stage YOLO with CNN head for a bakery inspection pipeline (1M+ pieces/day), achieving < 100 ms inference latency and cutting \$60K in outsourcing costs.
- Engineered an internal computer vision tooling suite deployed at 2 customer sites and adopted company-wide across the engineering team, improving model evaluation and deployment workflows.
- Designed custom calibration tools for 6-DOF ABB robotic arm and vision cells in high-speed automated bread-scoring systems.

Research Assistant | MARS Lab

May 2025 – May 2026

- Designed a custom **4-bar linkage folding mechanism** and 10+ 3D-printed mechanical components (SLS nylon and carbon fiber) to maximize stiffness-to-weight ratio and increase payload capacity by 20%.
- Formulated forward/inverse kinematics for quadruped locomotion and built an **end-to-end Gazebo simulation pipeline** to validate motor torque and gait behavior.
- Implemented OctoMap-based 3D mapping, navigation, and obstacle avoidance with PX4 & Jetson Nano via MAVROS.

Design Team Lead | Ontario Tech RoboMaster

September 2024 – Present

- Directed the mechanical design of a 500+ component competition robot in SolidWorks with **full DFM/DFA compliance** for CNC machining, sheet metal, and 3D printing.
- Engineered a 2-DOF turret with belts and herringbone gears, reducing backlash and improving the accuracy of the shooter by 27%.
- Performed **failure analysis and FEA**, managed CAD version control, standardized team processes for seamless design across subsystems.
- Led a team of 6+ team members in mechanical design, **GD&T**, engineering drawings, and robotics design principles.

## PROJECTS

SensQ – Autonomous Mobile Robot Platform

January 2026 – Present

- Designed and 3D-printed an AMR with a custom suspension mechanism, achieving speeds of 0.7 m/s with sub-2mm repeatability in point-to-point navigation tasks.
- Simulated system performance in Gazebo and integrated end-to-end physical hardware with ROS 2.
- Implemented Nav2 and SLAM Toolbox navigation, custom A\* and Boustrophedon coverage planning, and AprilTag-based docking.

Bolt Eye – Automated Inspection System

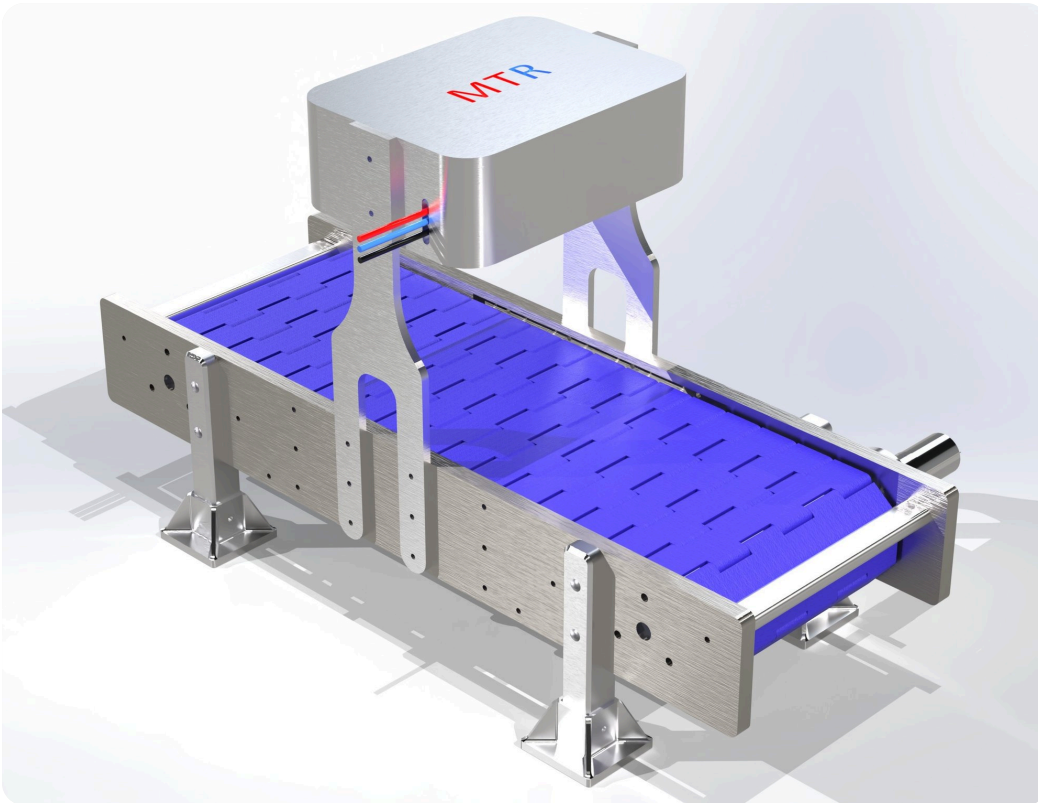
August 2025 – Present

- Built a 1 m/s motorized conveyor with encoder tracking, belt tensioner, and synchronized vision acquisition for real-time inspection.
- Developed a real-time defect detection and classification pipeline using YOLOv11 and DINOv2, achieving sub-100 ms inference latency.
- Detected and classified surface defects as small as 2 mm on an active production-line conveyor operating at 1 m/s.

6-DOF Robotic Arm

May 2024 – January 2026

- Calculated joint torque profiles, gear reductions, and preload compensation across 6 degrees of freedom, increasing structural payload capacity to 2 kg while maintaining structural rigidity.
- Customized kinematics calculation with L-BFGS-B path planning for pick and place tasks with 2mm accuracy.
- Implemented with Python, MoveIt, and RViz tools for motion visualization and validation.



# Bolt Eye

## Automated Inspection System

A conveyor vision cell that counts, sorts and quality-checks screws, bolts and nuts as they pass.

### HIGHLIGHTS

- End-to-end design, fully 3D-printed.
- Real-time detection and segmentation.
- Sorts parts by size and shape with millimetre accuracy.

### SKILLS

Computer vision

3D printing

Real-time systems

CAD design

### TOOLS

YOLOv11

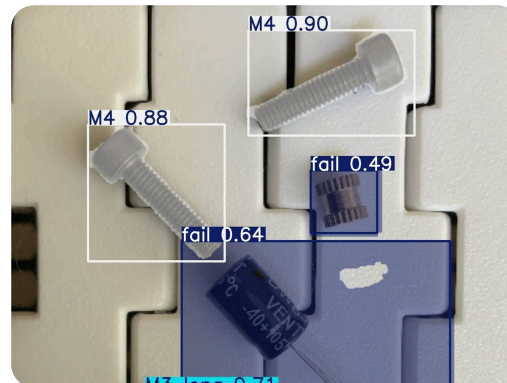
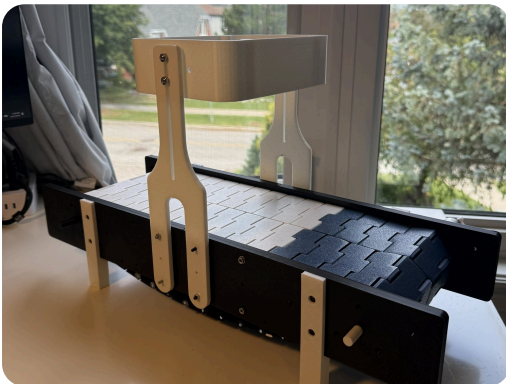
DINOv2

OpenCV

PyTorch

SolidWorks

Jetson Nano



# RoboMaster Robot

## Omnidirectional Chassis and Turret

A robot built for combat in the RoboMaster North America (ARC) competition.

### HIGHLIGHTS

- Mechanical design directed end to end in SolidWorks, DFM/DFA compliant.
- Custom turret: timing-belt yaw and herringbone-gear pitch, minimizing backlash for accuracy.

### SKILLS

Mechanical design

GD&amp;T

Design for manufacture

Design leadership

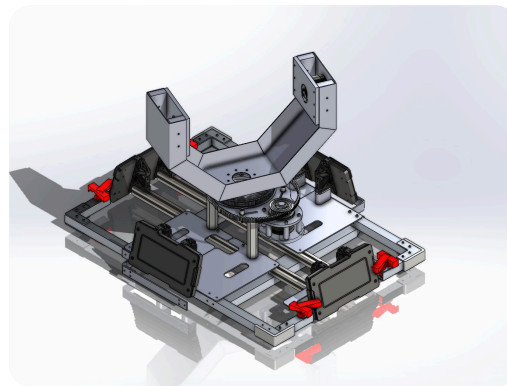
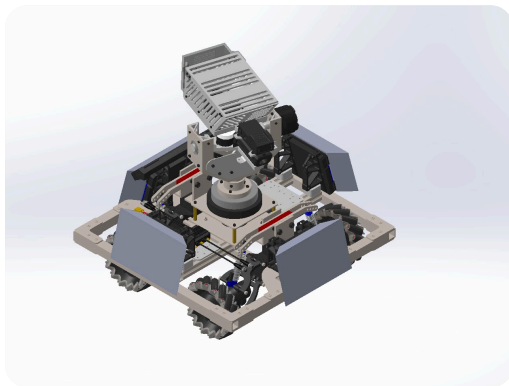
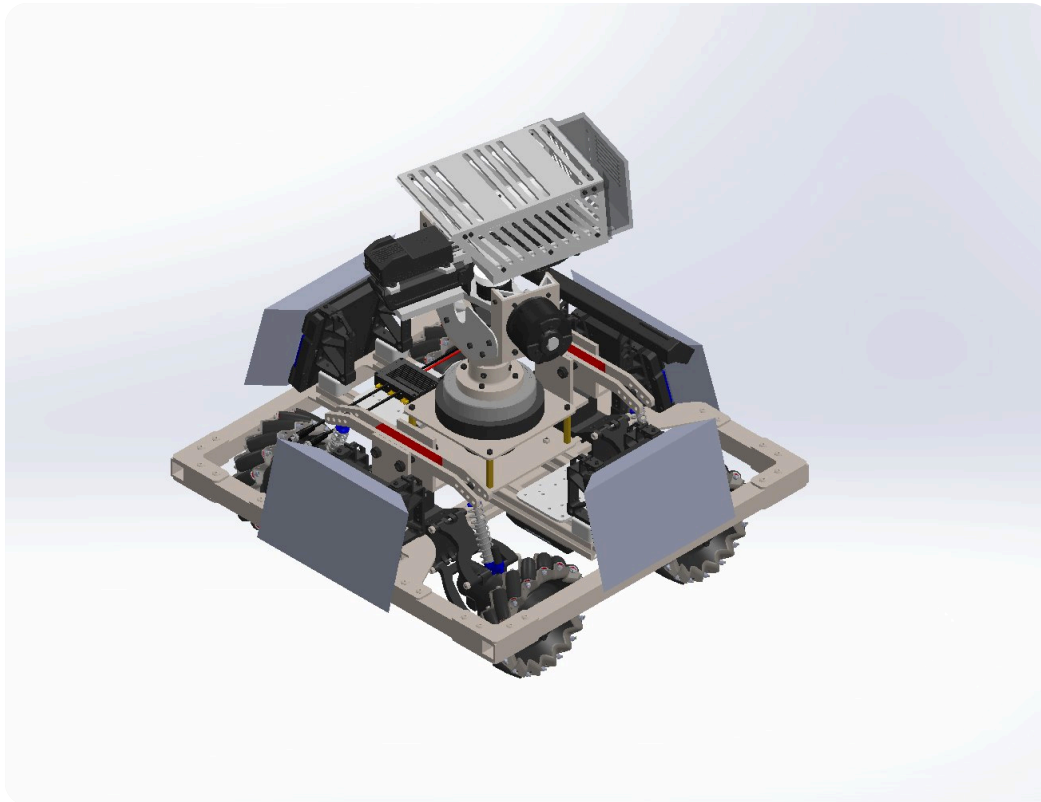
### TOOLS

SolidWorks

GD&amp;T

DFM/DFA

Engineering drawings



# Digital Twin Work Cell

## Isaac Sim Validation Pipeline

A virtual copy of a robotic bread-scoring cell, so layouts and motions are proven before any hardware moves.

### HIGHLIGHTS

- Arm, conveyor and sensors simulated closely to the real cell.
- Robot motion and system behavior tested entirely in simulation.
- Significantly reduced commissioning time.

### SKILLS

Robotics simulation

Industrial communication protocols

### TOOLS

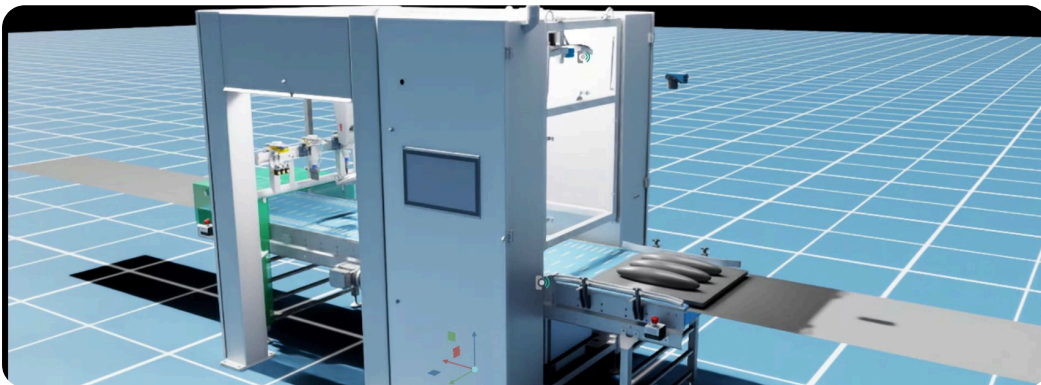
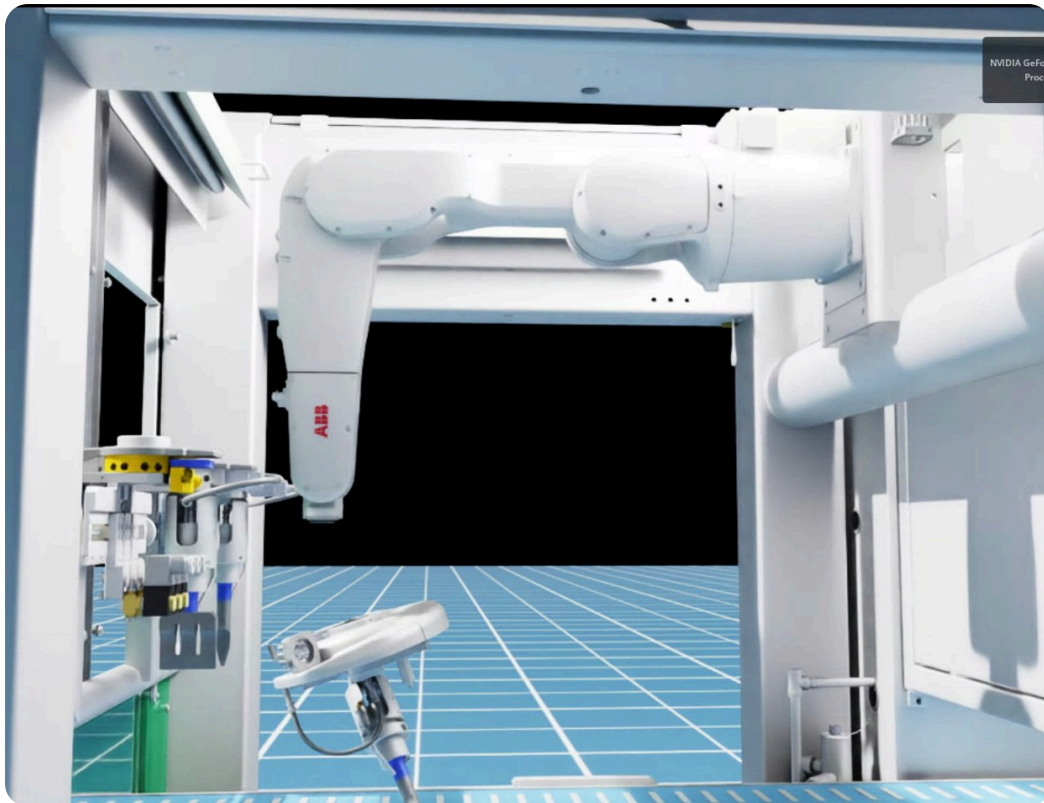
NVIDIA Isaac Sim

RobotStudio

ABB IRB1200

ROS 2

Python





# Pegasus

## Hybrid Multimodal Quadruped

A research robot that combines walking and flying on one platform.

### HIGHLIGHTS

- Increased payload over the original robot by 15% with custom trot kinematics.
- Designed custom sensor mounts.
- Simulated in Gazebo for behavior validation.

### SKILLS

Legged locomotion

Mechanical design

Kinematics

3D mapping

### TOOLS

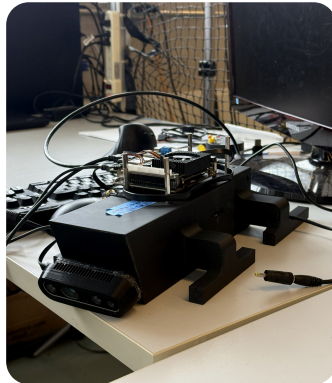
ROS

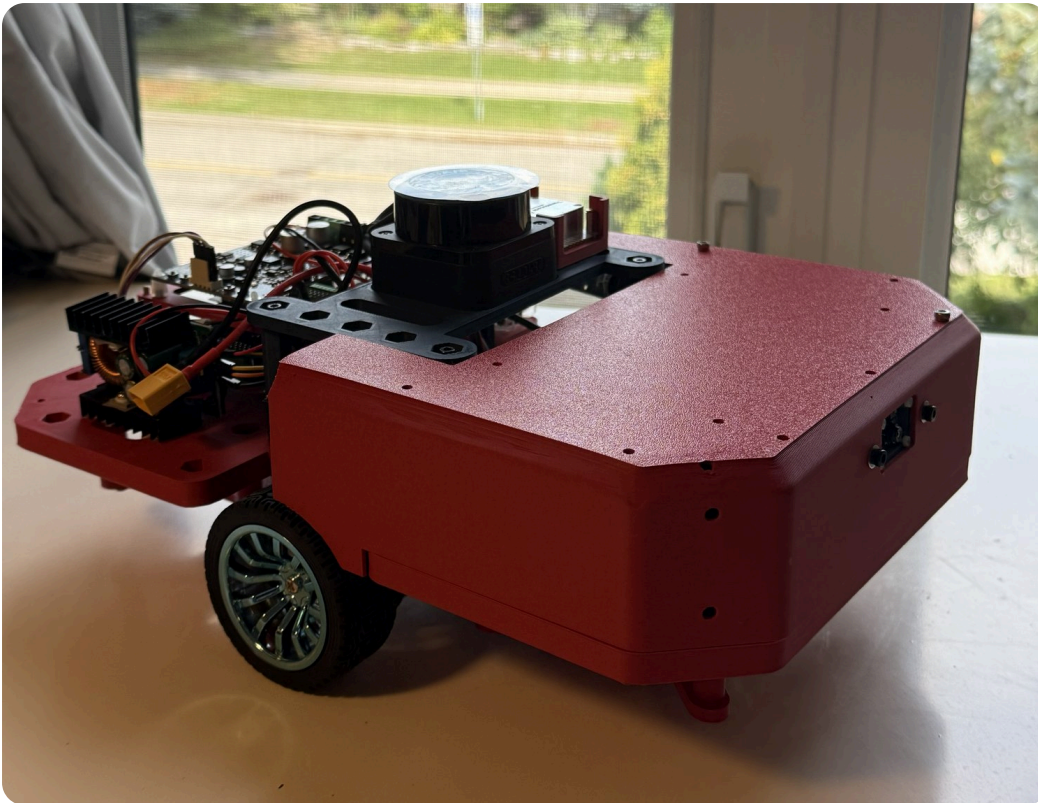
Gazebo

PX4/MAVROS

OctoMap

SolidWorks





# SensQ

## Autonomous Mobile Robot Platform

A compact robot that maps a space, covers every part of it, and finds its own way back to dock.

### HIGHLIGHTS

- Chassis and hardware 3D-printed and assembled in-house.
- Coverage path planning with A\* and Boustrophedon cell decomposition.
- Docking and alignment with AprilTags.

### SKILLS

Mobile robotics

SLAM

Path planning

Hardware integration

### TOOLS

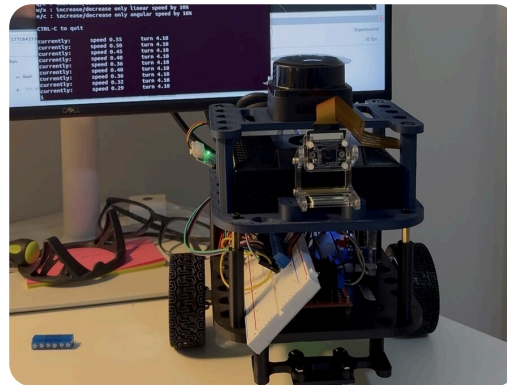
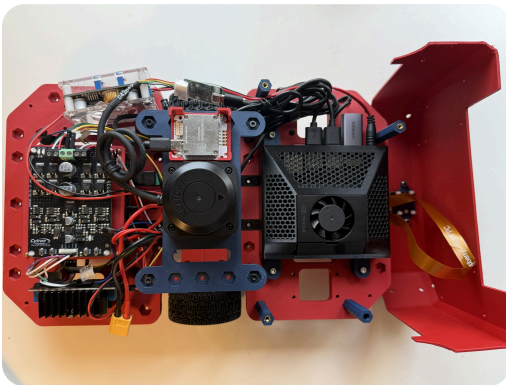
ROS 2

Nav2

SLAM Toolbox

AprilTag

Python



# 6-DOF Robotic Arm

## Pick and Place Robot

A six-axis robot arm for pick-and-place tasks with high precision and repeatability.

### HIGHLIGHTS

- Designed in SolidWorks and fully 3D-printed, with a 2 kg payload and minimal backlash.
- Path planning with L-BFGS-B optimization for smooth, reachable motion.
- Simulated in Gazebo to preview motion.

### SKILLS

Robot kinematics

Optimization

Motion planning

CAD design

### TOOLS

ROS

RViz

Python

L-BFGS-B

SolidWorks

